

UNIT 4 – LAB EXPERIMENTS

COURSE CODE: CSA1610

COURSE NAME: DWDM

G.Hima Bindu(192511377)

EXPERIMENT NO. 1

1.Consider that you are owning a supermarket mall and through membership cards, you have some basic data about your customers like Customer ID, age, gender, annual income and spending score. For the above scenario, the Problem Statement was You want to understand the customers who can easily converge [Target Customers] so that the data can be given to the marketing team and plan the strategy accordingly. For the above scenario prepare a dataset and perform **Clustering Analysis** to segment the customers in the Mall. There are clearly Five segments of Customers based on their Annual Income and Spending Score namely *Usual Customers, Priority Customers, Senior Citizen Target Customers, and Young Target Customers*.Sample data

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

@relation mall_customer_segmentation

@attribute CustomerID numeric

@attribute Gender {Male,Female}

@attribute Age numeric

@attribute AnnualIncome numeric

@attribute SpendingScore numeric

@data

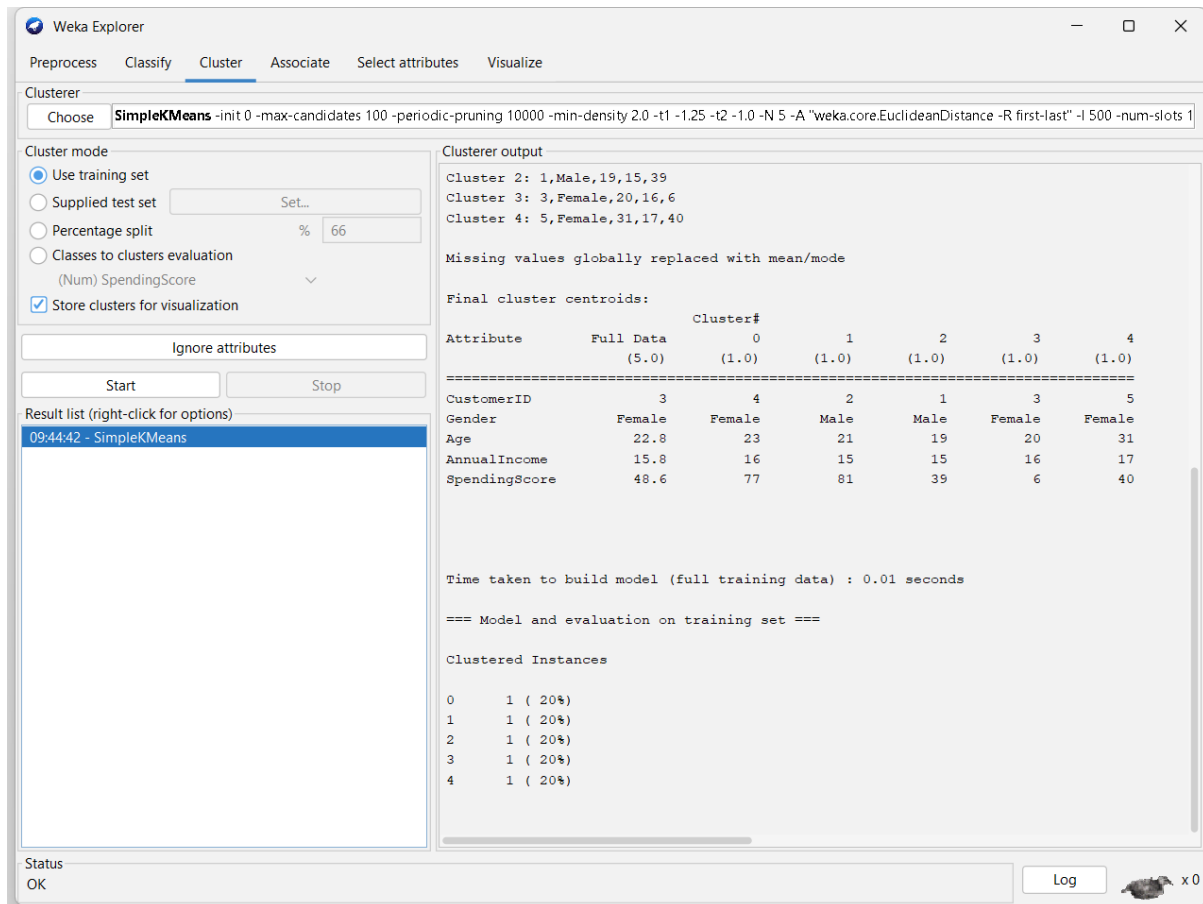
1, Male, 19, 15, 39

2, Male, 21, 15, 81

3, Female, 20, 16, 6

4,Female,23,16,77

5,Female,31,17,40



EXPERIMENT NO.: 2

2.Create the following dataset using CSV file format. To perform cluster analysis using K-Means in WEKA. To change the cluster size and plot the graph and illustrate the visualization of cluster.

EmployeeID	Gender	Age	Salary	Credit
111	Male	28	150000	39
222	Male	25	150000	27
333	Female	26	160000	42
444	Female	25	160000	40
555	Female	30	170000	64
666	Male	29	200000	72

@relation employee_cluster_analysis

@attribute EmployeeID numeric

@attribute Gender {Male,Female}

@attribute Age numeric

@attribute Salary numeric

@attribute Credit numeric

@data

111,Male,28,150000,39

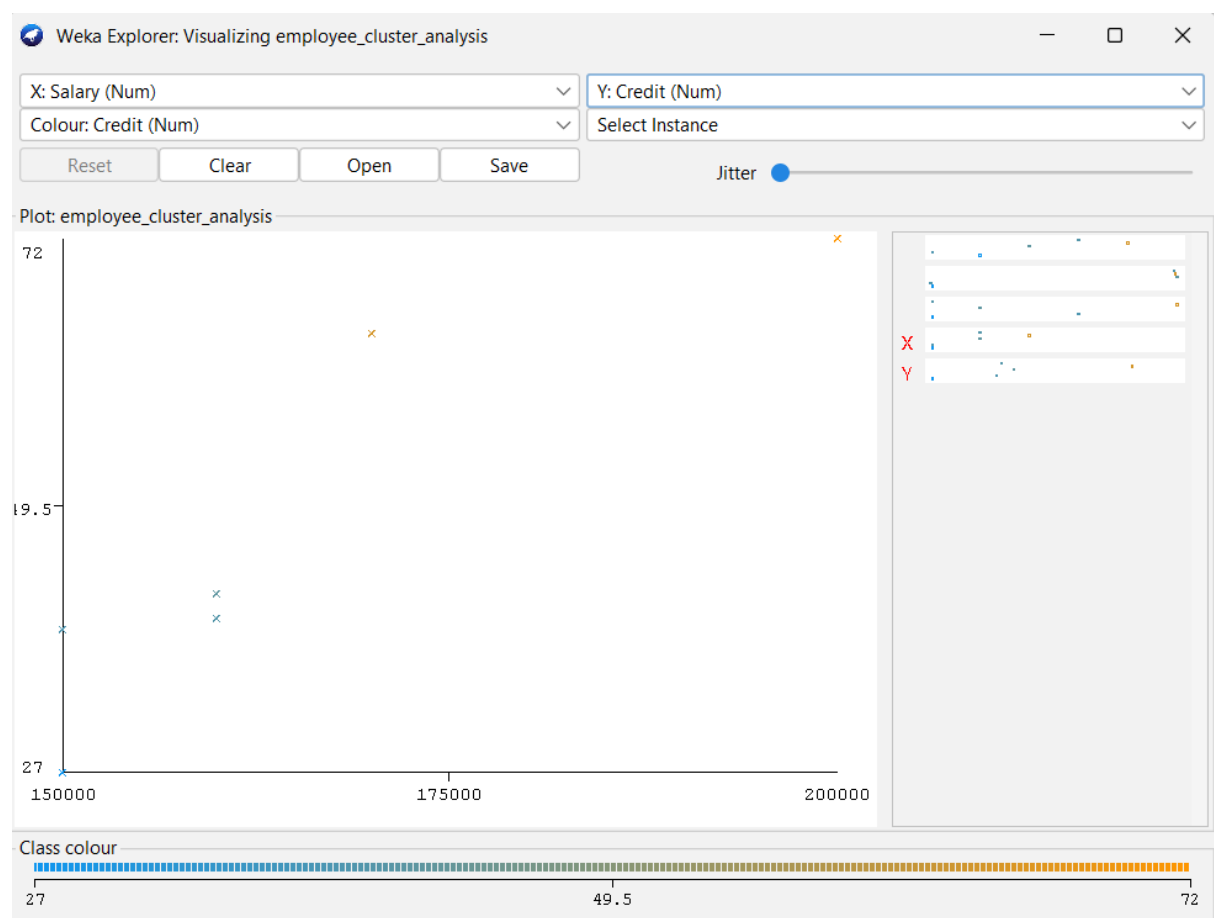
222,Male,25,150000,27

333,Female,26,160000,42

444,Female,25,160000,40

555,Female,30,170000,64

666,Male,29,200000,72



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation
(Num) Credit v

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

09:44:42 - SimpleKMeans

09:47:14 - SimpleKMeans

Cluster output

Cluster 2: 222, Male, 25, 150000, 27

Cluster 3: 555, Female, 30, 170000, 64

Cluster 4: 333, Female, 26, 160000, 42

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (6.0)	Cluster# 0 (2.0)	1 (1.0)	2 (1.0)	3 (1.0)	4 (1.0)
EmployeeID	388.5	388.5	111	222	666	555
Gender	Male	Female	Male	Male	Male	Female
Age	27.1667	25.5	28	25	29	30
Salary	165000	160000	150000	150000	200000	170000
Credit	47.3333	41	39	27	72	64

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (33%)
1	1 (17%)
2	1 (17%)
3	1 (17%)
4	1 (17%)

Status
OK

Log x 0

EXPERIMENT NO.: 3

3. Prediction of categorical data using Naïve Bayes classification through WEKA using any datasets. Compare the Naïve Bayes algorithm with SVM using the summary of results given by the classifiers and plot the graph.

@relation student_performance

@attribute StudyHours {low,medium,high}

@attribute Attendance {poor,average,good}

@attribute Result {pass,fail}

@data

low,poor,fail

low,average,fail

medium,average,pass

medium,good,pass

high,good,pass

high,average,pass

low,good,fail

medium,poor,fail

high,poor,pass

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose NaiveBayes

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Result

Start Stop

Result list (right-click for options)

09:51:04 - bayes.NaiveBayes

Classifier output

Scheme: weka.classifiers.bayes.NaiveBayes

Relation: student_performance

Instances: 9

Attributes: 3

StudyHours

Attendance

Result

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

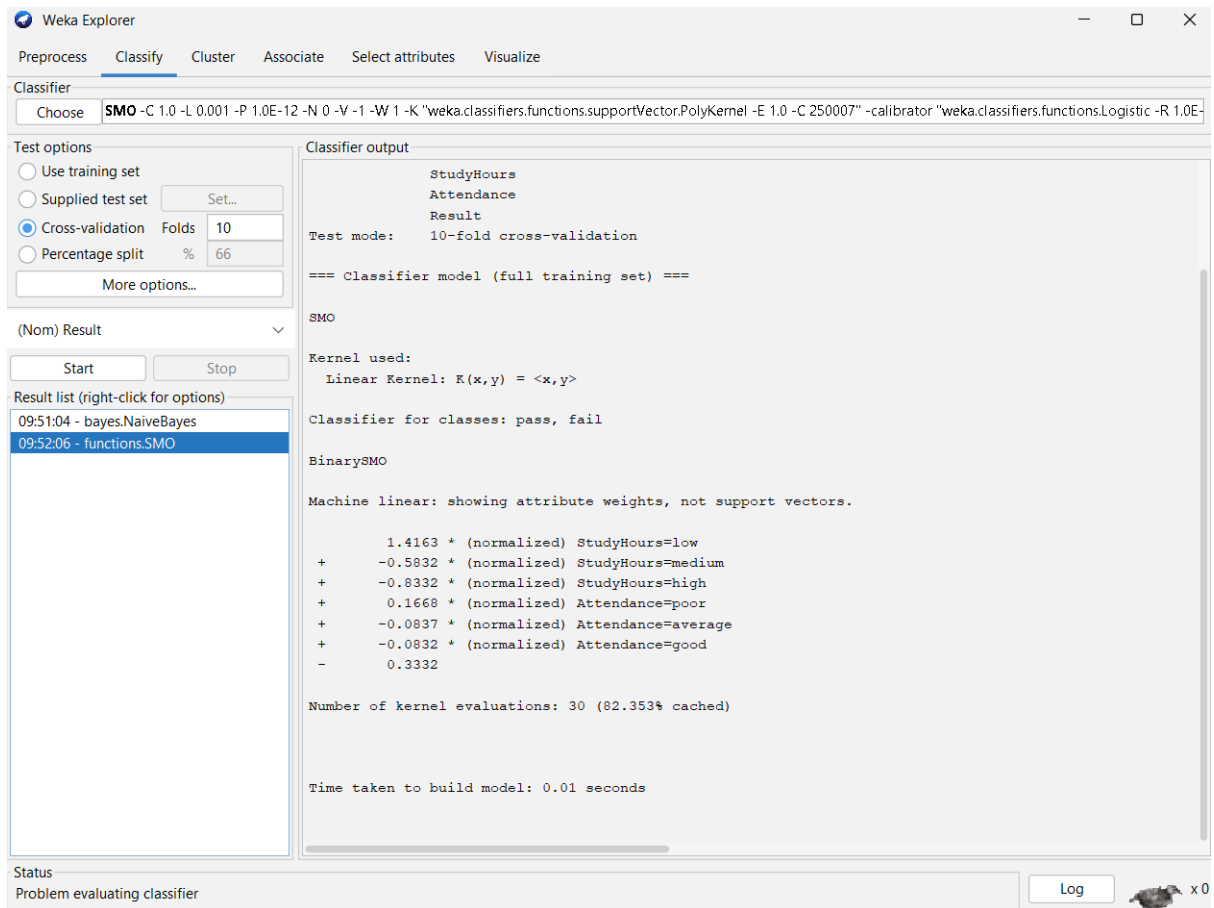
Attribute	Class	
	pass	fail
	(0.55)	(0.45)
=====		
StudyHours		
low	1.0	4.0
medium	3.0	2.0
high	4.0	1.0
[total]	8.0	7.0
Attendance		
poor	2.0	3.0
average	3.0	2.0
good	3.0	2.0
[total]	8.0	7.0

Time taken to build model: 0 seconds

Status

Problem evaluating classifier

Log x 0



EXPERIMENT NO.: 4

4.The following list of persons with vegetarian or not details given in the table. How will you find out how many of them are vegetarian and how many of them are non-vegetarian? Which type of the person total count is greater value?

Person	Gopu	Babu	Baby	Gopal	Krishna	Ja	De	Malin	Hema	Anu
Vegetarian	yes	yes	yes	no	yes	no	no	yes	yes	yes

@relation vegetarian_analysis

@attribute Person string

@attribute Vegetarian {yes,no}

@data

Gopu,yes

Babu,yes

Baby,yes

Gopal,no

Krishna,yes

Jai,no

Dev,no

Malini,yes

Hema,yes

Anu,yes

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose ZeroR

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Vegetarian

Start Stop

Result list (right-click for options)

- 09:51:04 - bayes.NaiveBayes
- 09:52:06 - functions.SMO
- 09:54:54 - rules.ZeroR

Classifier output

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

ZeroR predicts class value: yes

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	7	70	%
Incorrectly Classified Instances	3	30	%
Kappa statistic	0		
Mean absolute error	0.4727		
Root mean squared error	0.5012		
Relative absolute error	100	%	
Root relative squared error	100	%	
Total Number of Instances	10		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	1.000	0.700	1.000	0.824	?	0.000	0.700	yes
	0.000	0.000	?	0.000	?	?	0.000	0.300	no
Weighted Avg.	0.700	0.700	?	0.700	?	?	0.000	0.580	

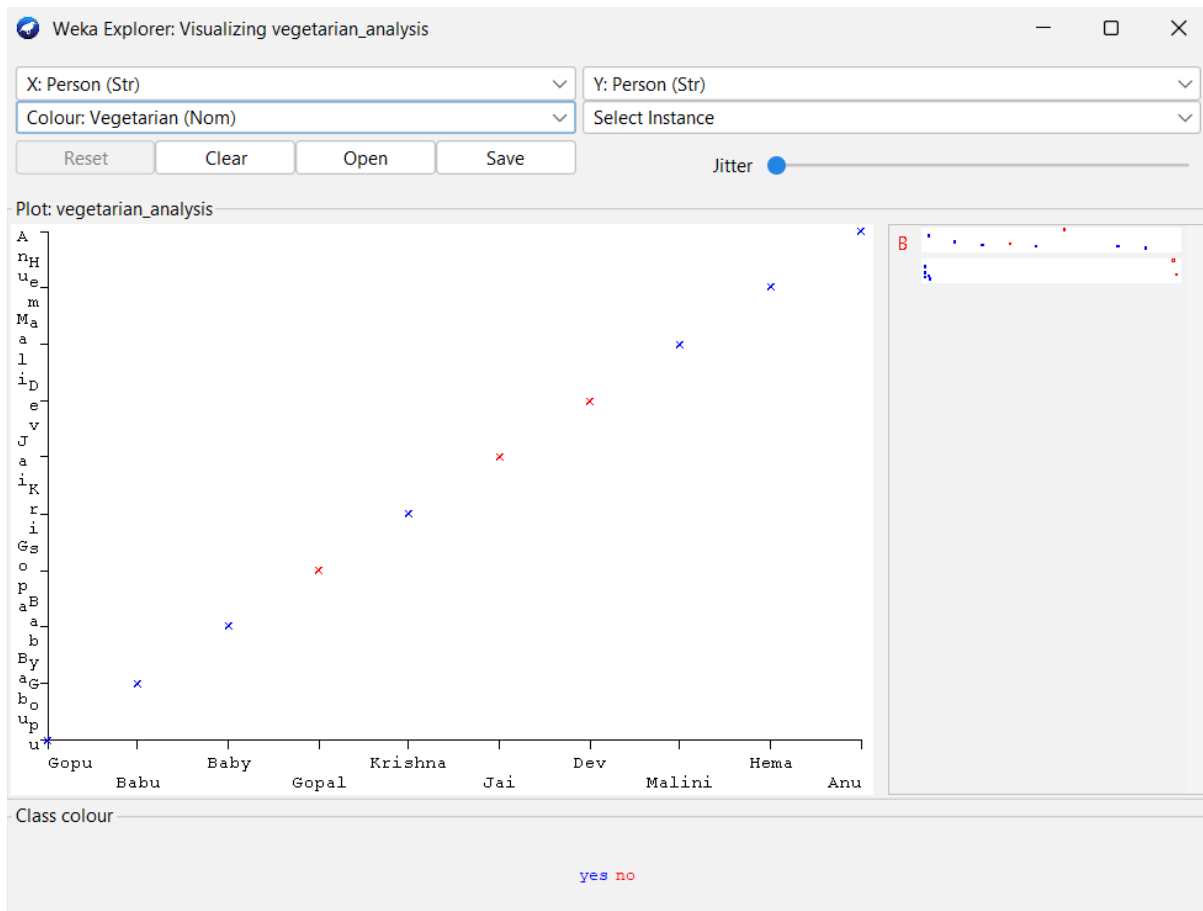
=== Confusion Matrix ===

```
a b <-- classified as
7 0 | a = yes
3 0 | b = no
```

Status

OK

Log x0



EXPERIMENT NO.: 5

5.The following table would be plotted as (x,y) points, with the first column being the x values as number of mobile phones sold and the second column being the y values as money. To use the scatter plot for how many mobile phones sold.

x	4	1	5	7	10	2	50	25	90	36
y	12	5	13	19	31	7	153	72	275	110

@relation mobile_sales

@attribute PhonesSold numeric

@attribute Money numeric

@data

4,12

1,5

5,13

7,19

10,31

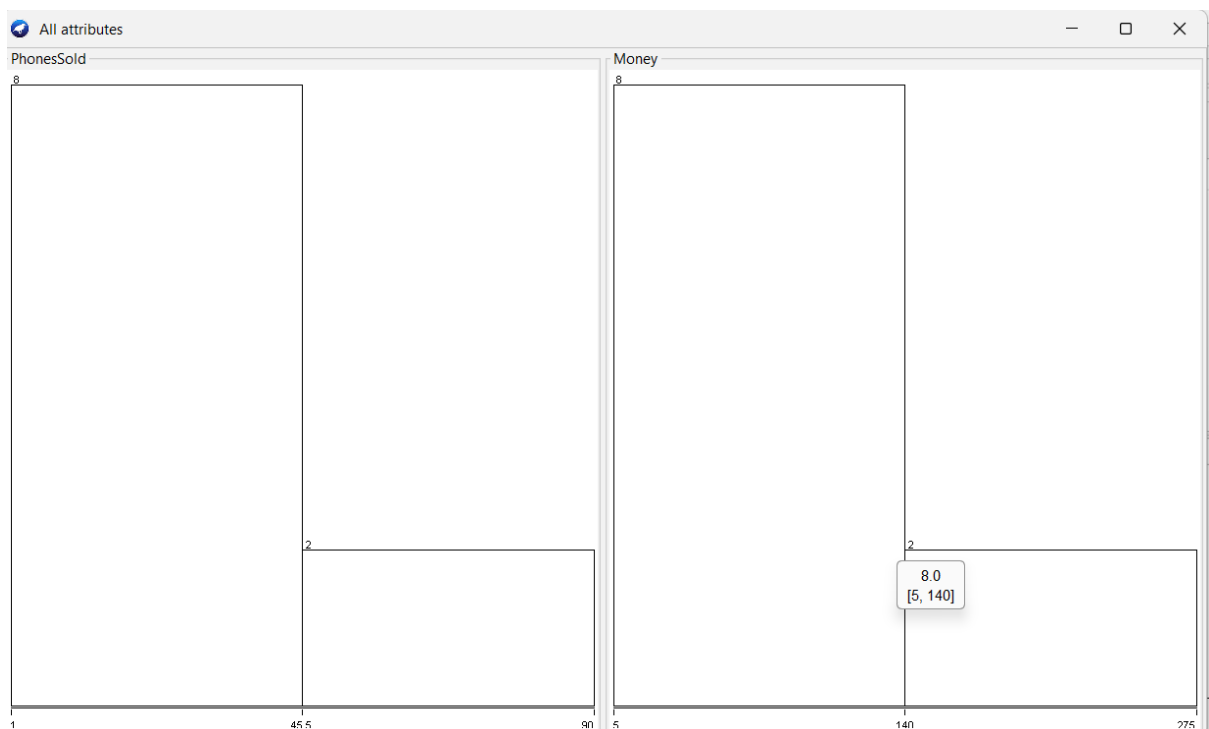
2,7

50,153

25,72

90,275

36,110



EXPERIMENT NO.: 6

6. Generate rules using FP growth algorithm using the given dataset which has the following transactions with items purchased: Consider the values as support=50% and confidence=75%.

Transaction ID	Items Purchased
1	Bread, Cheese, Egg, Juice
2	Bread, Cheese, Juice
3	Bread, Milk, Yogurt
4	Bread, Juice, Milk
5	Cheese, Juice, Milk

@relation items

@attribute Bread {t,f}

@attribute Cheese {t,f}

@attribute Egg {t,f}

@attribute Juice {t,f}

@attribute Milk {t,f}

@attribute Yogurt {t,f}

@data

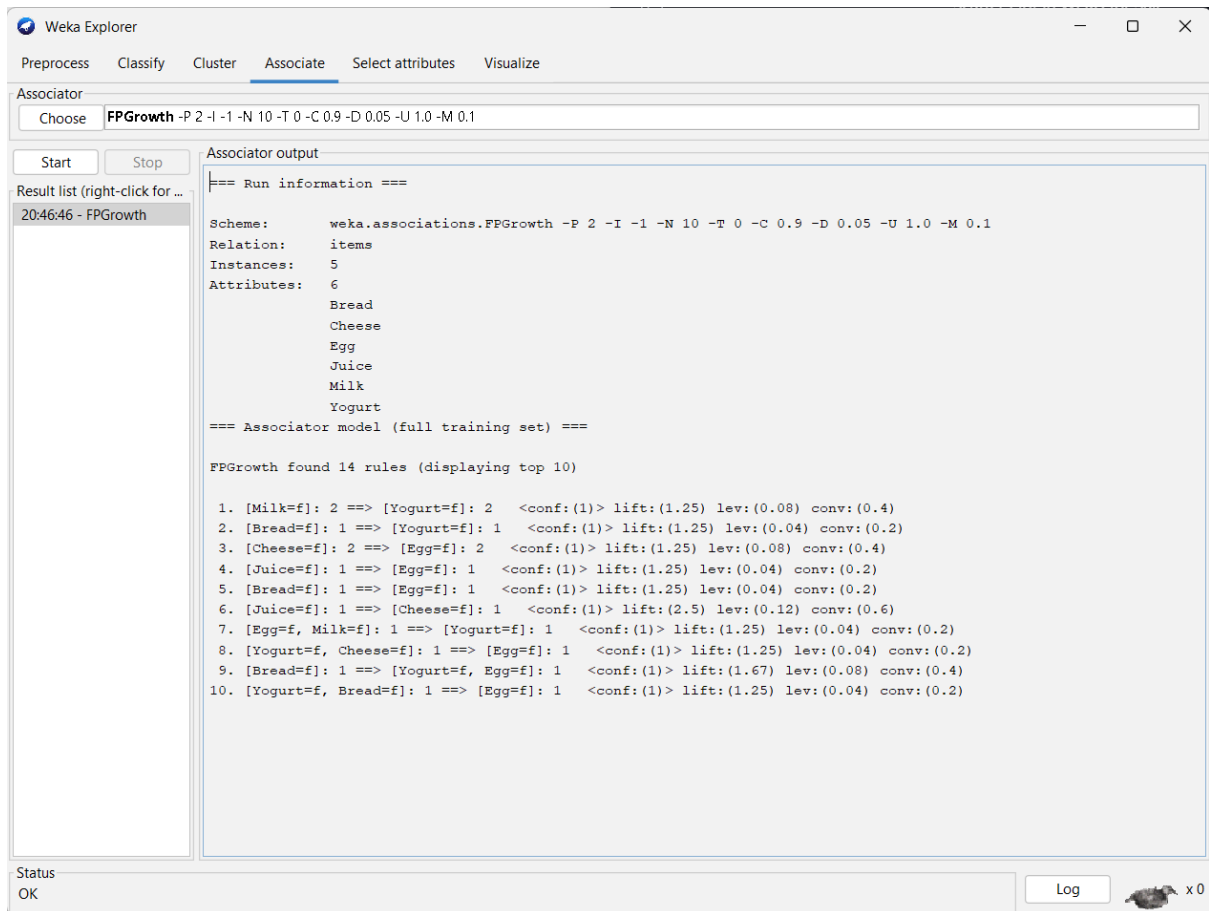
t,t,t,t,f,f

t,t,f,t,f,f

t,f,f,f,t,t

t,f,f,t,t,f

f,t,f,t,t,f



EXPERIMENT NO.:7

@relation diabetes

@attribute Pregnancies numeric

@attribute Glucose numeric

@attribute BloodPressure numeric

@attribute SkinThickness numeric

@attribute Insulin numeric

@attribute BMI numeric

@attribute DiabetesPedigreeFunction numeric

@attribute Age numeric

@attribute Outcome {tested_positive, tested_negative}

@data

6,148,72,35,0,33.6,0.627,50,tested_positive
 1,85,66,29,0,26.6,0.351,31,tested_negative
 8,183,64,0,0,23.3,0.672,32,tested_positive
 1,89,66,23,94,28.1,0.167,21,tested_negative
 0,137,40,35,168,43.1,2.288,33,tested_positive
 5,116,74,0,0,25.6,0.201,30,tested_negative
 3,78,50,32,88,31.0,0.248,26,tested_positive
 10,115,0,0,0,35.3,0.134,29,tested_negative
 2,197,70,45,543,30.5,0.158,53,tested_positive
 8,125,96,0,0,0.0,0.232,54,tested_negative

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier
 Choose **J48 -C 0.25 -M 2**

Test options
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) Outcome
 Start Stop

Result list (right-click for options)
 21:41:38 - trees.J48

Classifier output

Number of Leaves : 2
 Size of the tree : 3
 Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
 === Summary ===

Correctly Classified Instances	7	70	%
Incorrectly Classified Instances	3	30	%
Kappa statistic	0.4		
Mean absolute error	0.38		
Root mean squared error	0.5621		
Relative absolute error	69.6667	%	
Root relative squared error	103.0588	%	
Total Number of Instances	10		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.600	0.200	0.750	0.600	0.667	0.408	0.540	0.650	tested_p
	0.800	0.400	0.667	0.800	0.727	0.408	0.540	0.633	tested_n
Weighted Avg.	0.700	0.300	0.708	0.700	0.697	0.408	0.540	0.642	

=== Confusion Matrix ===

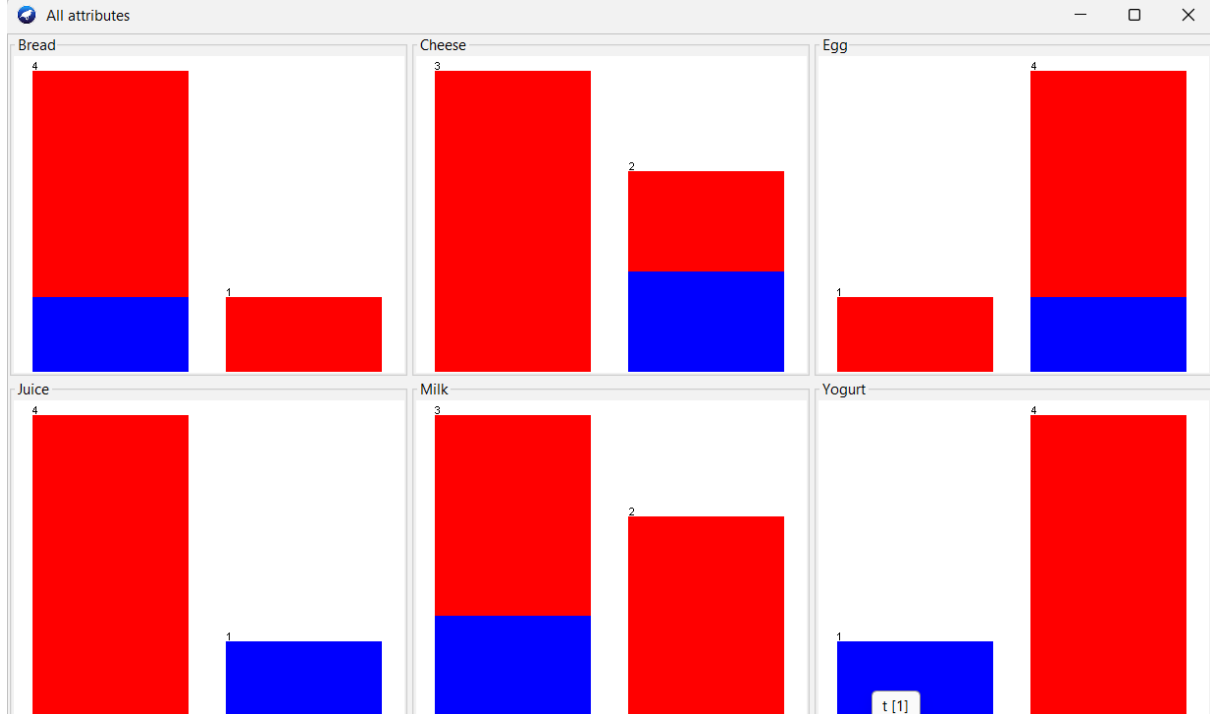
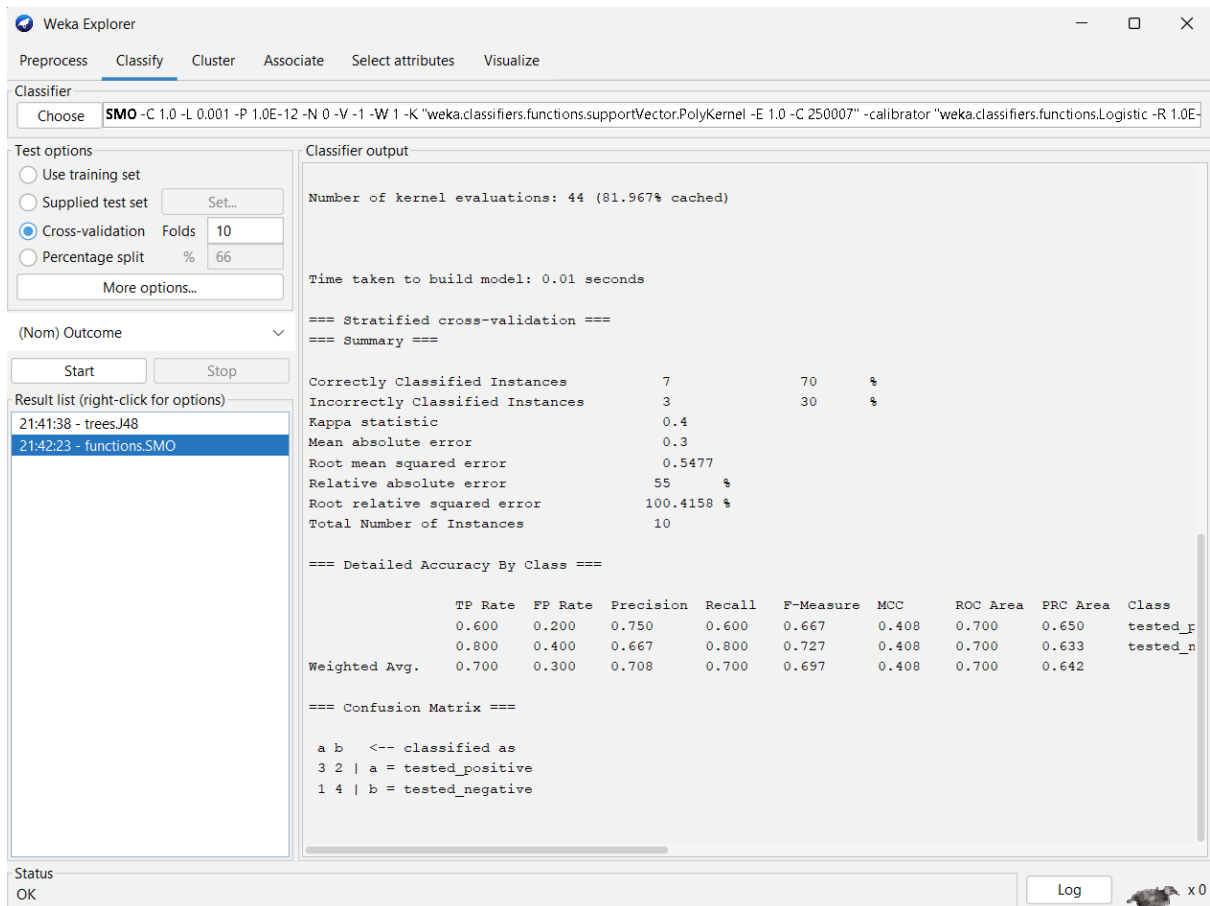
```

a b  <-- classified as
3 2 | a = tested_positive
1 4 | b = tested_negative

```

Status
 OK

Log



EXPERIMENT NO.:8

@relation marks_partition

@attribute Marks numeric

@data

55

60

71

63

55

65

50

55

58

59

61

63

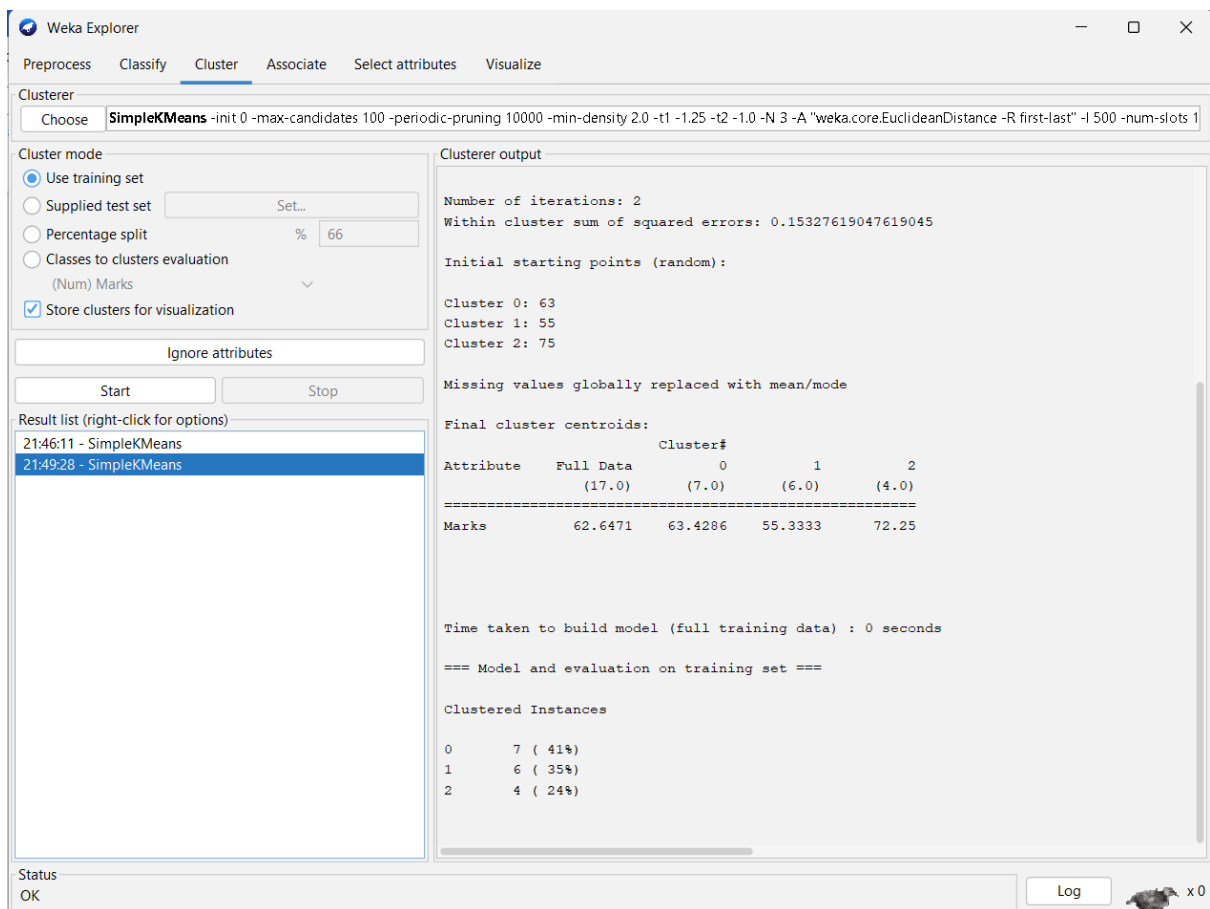
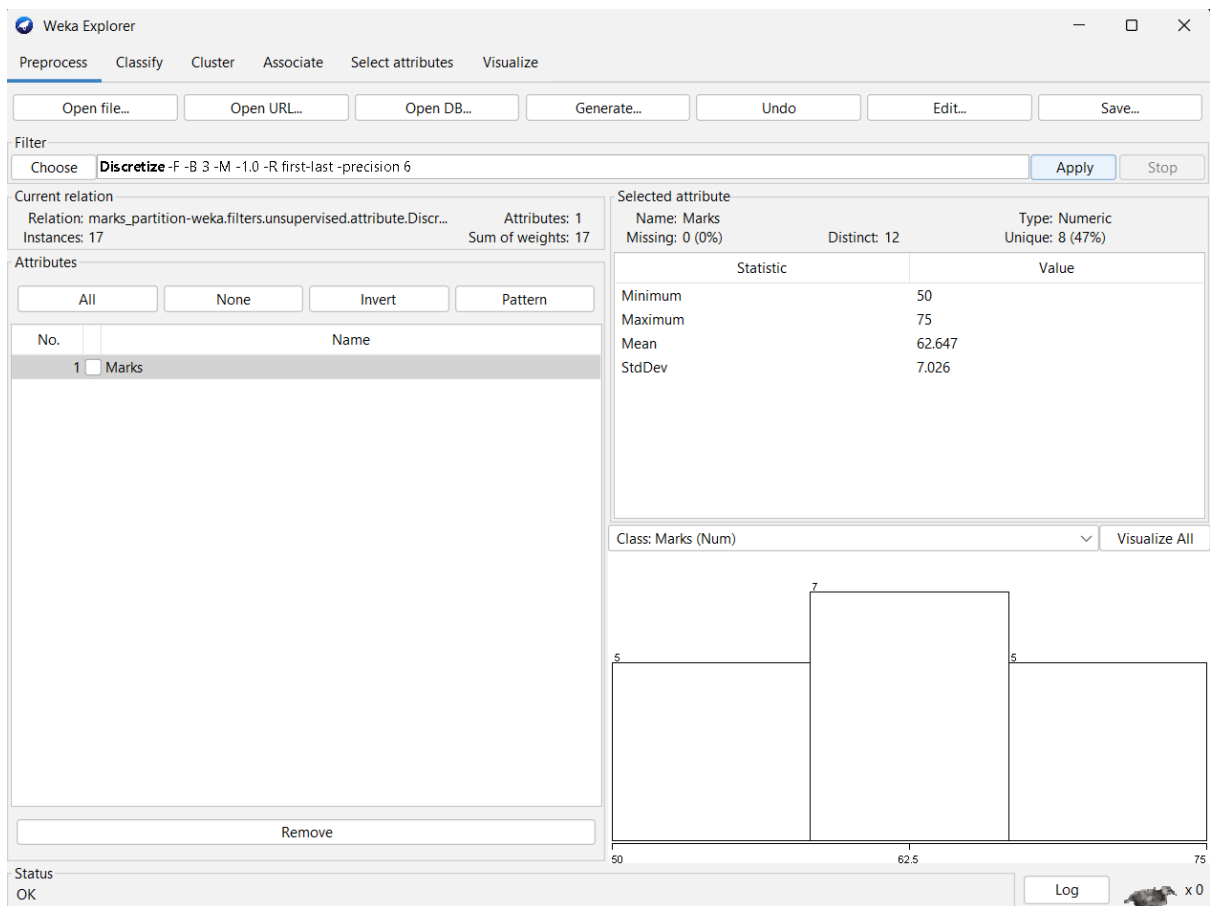
65

67

71

72

75



EXPERIMENT NO.:9

@relation gender_decision

@attribute Height numeric

@attribute Weight numeric

@attribute Gender {Male,Female}

@data

185,70,Male

190,85,Male

175,90,Male

160,55,Female

170,60,Female

165,75,Female

182,82,Male

178,78,Female

185,95,Male

172,65,Female

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds
- ☐ Percentage split %

(Nom) Gender

Result list (right-click for options):

- 21:51:39 - trees.J48

Classifier output:

Number of Leaves : 2

Size of the tree : 3

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	7	70	%
Incorrectly Classified Instances	3	30	%
Kappa statistic	0.4		
Mean absolute error	0.44		
Root mean squared error	0.5727		
Relative absolute error	80.6667	%	
Root relative squared error	104.9974	%	
Total Number of Instances	10		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.600	0.200	0.750	0.600	0.667	0.408	0.480	0.650	Male
	0.800	0.400	0.667	0.800	0.727	0.408	0.480	0.633	Female
Weighted Avg.	0.700	0.300	0.708	0.700	0.697	0.408	0.480	0.642	

=== Confusion Matrix ===

```

a b  <-- classified as
3 2 | a = Male
1 4 | b = Female

```

Status: OK  x0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **JRip -F 3 -N 2.0 -O 2 -S 1**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds
- ☐ Percentage split %

(Nom) Gender

Result list (right-click for options):

- 21:51:39 - trees.J48
- 21:52:43 - rules.JRip

Classifier output:

(Height >= 182) => Gender=Male (4.0/0.0)
=> Gender=Female (6.0/1.0)

Number of Rules : 2

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	8	80	%
Incorrectly Classified Instances	2	20	%
Kappa statistic	0.6		
Mean absolute error	0.36		
Root mean squared error	0.4817		
Relative absolute error	66	%	
Root relative squared error	88.305	%	
Total Number of Instances	10		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.800	0.200	0.800	0.800	0.800	0.600	0.640	0.740	Male
	0.800	0.200	0.800	0.800	0.800	0.600	0.640	0.740	Female
Weighted Avg.	0.800	0.200	0.800	0.800	0.800	0.600	0.640	0.740	

=== Confusion Matrix ===

```

a b  <-- classified as
4 1 | a = Male
1 4 | b = Female

```

Status: OK  x0

EXPERIMENT NO.:10

@relation electronics_transactions

@attribute SONY {t,f}

@attribute BPL {t,f}

@attribute LG {t,f}

@attribute SAMSUNG {t,f}

@attribute ONIDA {t,f}

@data

t,t,t,f,f

f,t,f,t,f

f,t,f,f,t

t,t,f,t,f

t,f,f,f,t

f,t,f,f,t

t,f,f,f,t

t,t,t,f,t

t,t,f,f,t

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1**

Start Stop

Result list (right-click for ...)

21:56:08 - Apriori

Associator output

```

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.35 (3 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 13

Generated sets of large itemsets:

Size of set of large itemsets L(1): 7

Size of set of large itemsets L(2): 13

Size of set of large itemsets L(3): 9

Size of set of large itemsets L(4): 2

Best rules found:

1. ONIDA=t 6 ==> SAMSUNG=f 6 <conf:(1)> lift:(1.29) lev:(0.15) [1] conv:(1.33)
2. LG=f ONIDA=t 5 ==> SAMSUNG=f 5 <conf:(1)> lift:(1.29) lev:(0.12) [1] conv:(1.11)
3. LG=f SAMSUNG=f 5 ==> ONIDA=t 5 <conf:(1)> lift:(1.5) lev:(0.19) [1] conv:(1.67)
4. SONY=t ONIDA=t 4 ==> SAMSUNG=f 4 <conf:(1)> lift:(1.29) lev:(0.1) [0] conv:(0.89)
5. BPL=t ONIDA=t 4 ==> SAMSUNG=f 4 <conf:(1)> lift:(1.29) lev:(0.1) [0] conv:(0.89)
6. SONY=f 3 ==> BPL=t 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
7. SONY=f 3 ==> LG=f 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
8. ONIDA=f 3 ==> BPL=t 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
9. SONY=f LG=f 3 ==> BPL=t 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
10. SONY=f BPL=t 3 ==> LG=f 3 <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)

```

Status

OK

Log x0

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1**

Start Stop

Result list (right-click for ...)

21:56:08 - Apriori

21:56:34 - FPGrowth

Associator output

```

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation:    electronics_transactions
Instances:   9
Attributes:  5
             SONY
             BPL
             LG
             SAMSUNG
             ONIDA

=== Associator model (full training set) ===

FPGrowth found 8 rules (displaying top 8)

1. [BPL=f]: 2 ==> [SAMSUNG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
2. [SONY=f]: 3 ==> [LG=f]: 3 <conf:(1)> lift:(1.29) lev:(0.07) conv:(0.67)
3. [BPL=f]: 2 ==> [LG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
4. [SAMSUNG=f, SONY=f]: 2 ==> [LG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
5. [BPL=f]: 2 ==> [SAMSUNG=f, LG=f]: 2 <conf:(1)> lift:(1.8) lev:(0.1) conv:(0.89)
6. [SAMSUNG=f, BPL=f]: 2 ==> [LG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
7. [LG=f, BPL=f]: 2 ==> [SAMSUNG=f]: 2 <conf:(1)> lift:(1.29) lev:(0.05) conv:(0.44)
8. [SONY=f, ONIDA=f]: 1 ==> [LG=f]: 1 <conf:(1)> lift:(1.29) lev:(0.02) conv:(0.22)

```

Status

OK

Log x0

EXPERIMENT NO.:12

@relation strike_rate

@attribute StrikeRate numeric

@data

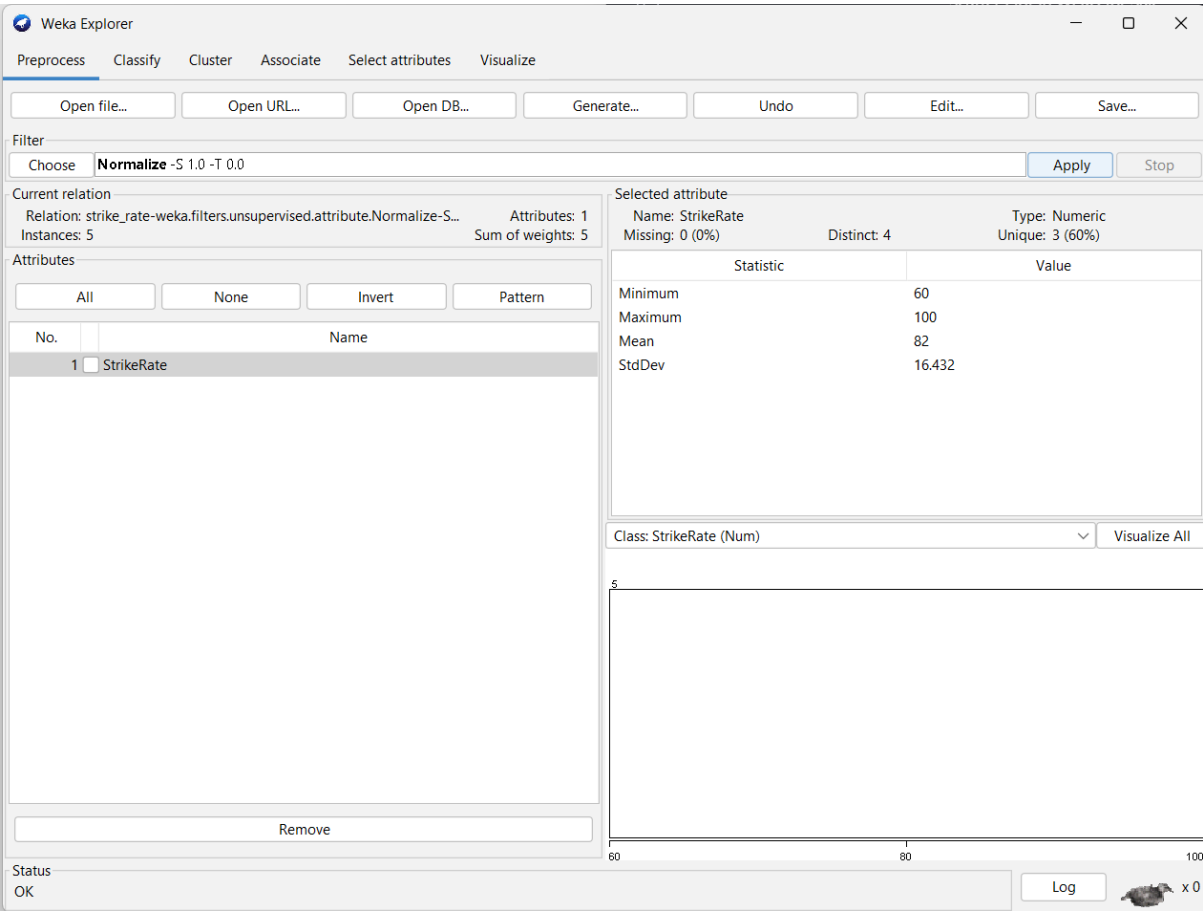
100

70

60

90

90



Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

Filter

ChooseStandardize

ApplyStop

Current relation

Relation: strike_rate-weka.filters.unsupervised.attribute.Normalize-S...
Instances: 5

Attributes: 1
Sum of weights: 5

Attributes

AllNoneInvertPattern

No.	Name
1	StrikeRate

Remove

Selected attribute

Name: StrikeRate
Missing: 0 (0%)
Distinct: 4
Type: Numeric
Unique: 3 (60%)

Statistic	Value
Minimum	60
Maximum	100
Mean	82
StdDev	16.432

Class: StrikeRate (Num)Visualize All

5

6080100

Status
OK

Log x 0